

SUMIT YADAV

Mechanistic Interpretability | AI Alignment & Safety | NLP

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RESEARCH STATEMENT

I work at the intersection of **mechanistic interpretability**, **AI alignment**, and **robustness**. My research studies how safety-aligned LLMs fail silently — through over-refusals, geometric misrepresentation, and surface-level triggers — and how the internal structure of representations can be understood and steered to make models safer and more reliable. Much of my strongest work is sole-authored: I combine representation-space analysis, adversarial probing, and systems engineering into empirically grounded methods. My work has been accepted at **ACL 2026 (CORE A*)**, and I am especially interested in extending interpretability and safety to multilingual and low-resource settings.

SELECTED RESEARCH & PUBLICATIONS

SafeConstellations: Mitigating Over-Refusals in LLMs Through Task-Aware Representation Steering | *ACL 2026 Main (CORE A*)* 2026

- LLMs trace consistent, task-specific “constellation” trajectories in embedding space that shift predictably between refusal and non-refusal states.
- **SafeConstellations** is an inference-time, task-aware trajectory-shifting method that nudges representations toward non-refusal pathways *only* on over-refusal-prone tasks — cutting over-refusals by up to **73%** with minimal utility loss and no retraining. (*with U. Maskey, M. Dras, U. Naseem.*)

Geometric Phases of Mechanism Formation in Neural Networks (*sole-authored*) | *GLOW 2026 @ IJCAI (Poster)* 2026

- Using linear probes and CKA across dense training checkpoints, shows classification mechanisms form **output-layer-first** and within the first ~5% of training (Cohen’s $d = 3.68$); the deep-first pattern holds in from-scratch LLM pretraining (GPT-2 Small, SmoLM2-135M) and reproduces on public Pythia and OLMo-2 checkpoints.

On the Relationship Between Representation Geometry and Generalization (*sole-authored*) | *Pre-Print 2026* 2026

- Establishes that a label-free geometric metric — **effective dimension** — strongly predicts generalization across vision and language (partial $r = 0.75$ over 52 ImageNet models, 13 architectures; carries to NLP encoders and decoder-only LLMs). The link is causal: degrading the geometry hurts accuracy while PCA cleanup preserves it — a diagnostic relevant to distribution-shift detection and robustness.

Steering Hindi Competence into Unsupported Devanagari Languages | *working paper* 2026

- With **no fine-tuning**, steers a model’s Hindi representations to elicit related Devanagari languages it was never trained on. On Llama-3.1-8B and Aya-23-8B a single steering direction produces **genuine Nepali**; Hindi outperforms English as the source because it sits closer in representation space, not because there is more of it. Unifies the SafeConstellations and maiBERT lines of work.

The Steerability Spectrum of Visual Attributes | *ongoing* 2026

- Measures how readily different visual features (size, distance, orientation) can be steered inside vision models (DINOv2, CLIP, SigLIP): some lie on a single clean, controllable direction while others are distributed and resist steering — probing what accounts for the difference.

MaiBERT: A Pre-training Corpus and Language Model for Low-Resourced Maithili | *LoResLM @ ACL 2026* 2026

- First monolingual BERT for Maithili (~50M speakers); **87.02%** news-classification accuracy, outperforming multilingual baselines (MuRIL, NepBERTa) — addressing fairness and access for underrepresented communities.

Also: **Revolutionizing Currency Security with YOLOv8** (J. Bus. Econ. Stud., 2024; 0.986 true-positive recall) · **Machine Learning Analysis of Tirhuta Lipi** (2023) · **Support Vectors for Imbalanced Text Classification** (2023).

EXPERIENCE

Astha.ai USA (Remote)
AI Researcher — Safety & Agentic Systems May 2025 – Present

- Built Zero-Trust agent oversight (cryptographic identity + policy engine on every tool call), an MCP security proxy (hexagonal architecture), and **MCP-Scanner** — a vulnerability platform integrating **78+** MITRE ATT&CK-mapped attack techniques for AI-agent infrastructure.

AMNIL Technologies Nepal (Hybrid)
AI Engineer — RAG & Infrastructure Jun 2024 – May 2025

- Guardrails + **LLM-as-a-Judge** evaluation; hybrid-search RAG (Qdrant, +35% retrieval accuracy); self-hosted LLM serving with vLLM (40% lower latency).

GradeUp Educations Kathmandu, Nepal
Data Team Lead Jan 2022 – Jun 2024

- Led the data team: learning agents/chatbots, an automated grade-evaluation system, and semantic-similarity matching (NLP + image processing).

DeepLearning.AI Global (Remote)
GAN Specialization Mentor Aug 2021 – Present

- Technical mentor for the GANs Specialization (loss functions, training stability, generative-model robustness).

SELECTED PROJECTS & OPEN SOURCE

SAFE-MCP — community security framework for the Model Context Protocol (ATT&CK-style threat/defense rules; prompt-injection & tool-poisoning modules).

Telebot-Claude-Bridge — permission-gated, human-in-the-loop remote control of the Claude Code CLI (inline allow/deny per tool call, audit logging).

Maithili speech — open Whisper-tiny **ASR** and MMS-TTS **TTS** for Maithili (live HF Spaces demos); **maiBERT** — first open BERT for Maithili (HF).

NPL Coders — founder of Nepal's national data-science competition community.

TECHNICAL PROFICIENCY

Interpretability & Alignment: representation & activation steering, mechanistic interpretability, linear probes / CKA, representation geometry, effective dimension, over-refusal mitigation.

LLMs & NLP: LLMs (Llama, Claude, GPT), fine-tuning (PEFT/LoRA), RAG, multilingual / low-resource NLP, tokenizer training, embedding alignment.

Agentic & Security: Model Context Protocol, multi-agent frameworks, MITRE ATT&CK mapping, prompt-injection & tool-poisoning defense, adversarial evaluation.

Tools: Python (expert), PyTorch, TensorFlow, vLLM, FastAPI, Docker, Kubernetes, vector DBs (Qdrant, ChromaDB), nnsight.

Research methods: empirical NLP evaluation, information-theoretic metrics, adversarial benchmarking, statistical analysis.

EDUCATION

Pulchowk Engineering College, IOE — Tribhuvan University

Kathmandu, Nepal

Bachelor of Computer Engineering

2019 – 2024

- B.E. thesis: “Evaluating Auto-Encoding Transformer Language Models for Maithili Text Classification” — my first Maithili masked language model, and the foundation for maiBERT.
- Key subjects: Artificial Intelligence, Data Structures & Algorithms, Distributed Systems, Big Data Technologies, Network Security, Compiler Design.

HONORS & AWARDS

Outstanding Reviewer, Mechanistic Interpretability Workshop (ICML 2026) · Winner, GritFeat AI Hackathon (2023) · 1st Runner-Up, Locus Dataverse (2023) · 1st Runner-Up, Docsumo DataRush (2022)

Best AI Project, DELTA 3.0 (2023) · Winner, IT-Meet Image Challenge (2022) · Winner, LogPoint Capture The Flag (2022)